

DEPARTMENT OF MATHEMATICS
QUESTION BANK (I SEM)
ACADEMIC YEAR: 2022-23 (ODD SEMESTER)

MODULE-4: INTEGRAL CALCULUS

Sub: Mathematics-I

Sub Code: 22MATM/S/E/C11

1. Evaluate : $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dy dx dz$

2. Evaluate : $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dz dy dx$

3. Evaluate : $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}}$

4. Evaluate : $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz dz dy dx$

5. Evaluate : $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx$

6. Evaluate by Changing the order of integration $\int_0^a \int_0^{2\sqrt{ax}} x^2 dy dx \quad a > 0$

7. Evaluate by Changing the order of integration $\int_0^1 \int_x^{\sqrt{x}} xy dy dx$

8. Evaluate by changing into polar coordinates $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy dx$

9. Find the volume of the tetrahedron bounded by the Planes: $x=0, y=0, z=0$,

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

10. Prove that $\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m + n}$

11. Prove that $\int_0^{\frac{\pi}{2}} \sqrt{\sin \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin \theta}} d\theta = \pi$

12. Evaluate $\int_0^1 x^{3/2} (1-x)^{1/2} dx$, by expressing in terms of beta and gamma functions.

13. A pyramid is bounded by three coordinate planes and the plane $x+2y+3z=6$ compute the volume by double integration.

14. Find the area bounded between the circle $x^2 + y^2 = a^2$ and the line $x + y = a$.

15. Find the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by double integration.